

Markups across the income distribution: measurement and implications

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BGGMP Reading Group

This paper

- Increasing average markups in US: *supply side* explanations
 - This paper: how changes in the *demand side* of the economy contribute to the rise in markups
- **Empirical part:** quantify how retail markups vary with household income and aggregate income, across cities, over time
- **Model:** study how retail markups change with income distribution
 - info friction: household retrieve price quotes from firms
 - endogenous search effort

Results preview

- Higher income households pay higher retail markups due to:
 - differences in markups paid for identical products
 - differences in basket composition
- Positive and large spillovers of others' incomes on markups
 - “macro” elasticity of markups to income (8-15%) > “micro” elasticity (3-4%)
- Model: increase in income and income dispersion account for 10-14pp increase in aggregate markup between 1950-2018
 - accelerates after 1980: income dispersion

Roadmap

- ① Empirical evidence
- ② Model: framework and calibration
- ③ Model implications:
 - spillovers across income groups
 - effects of inequality changes
 - markups across cities
 - how changes in income distribution 1950-2018 affect markups

Empirical evidence

Data

- **Consumer panel data:** NielsenIQ Homescan - 2007
 - purchases, date, store location, UPC, saving from coupons, demographic data
- **Wholesale costs:** PromoData Price-Track
 - Deal prices as measure of retailers' wholesale costs
 - Compute national average wholesale price for each UPC
- **Data match:** 67,000 UPCs purchased match to wholesale costs
 - 43% of transactions, 37% expenditures

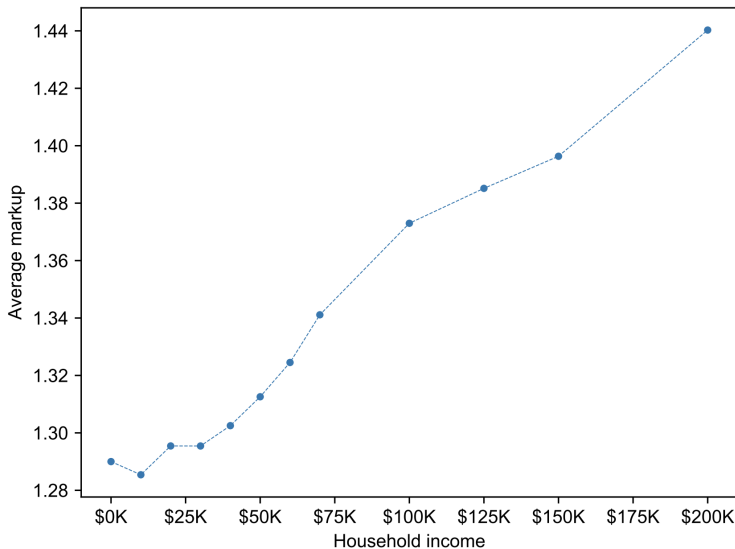
Retail markups estimates

- **Retail markups:**

$$\text{Retail markup}_{i,g,t} = \frac{\text{Price}_{i,g,t}}{\text{Wholesale cost}_{g,t}}$$

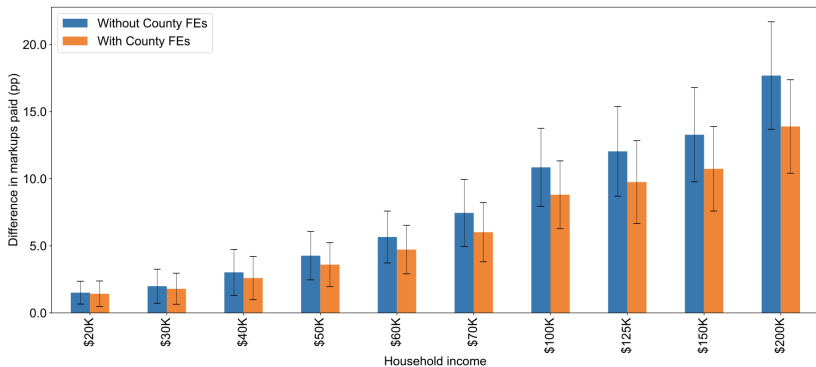
- Wholesale costs may not accurately measure retailers' marginal costs:
 - retailer-wholesaler deals
 - replacement costs $\neq$ wholesale costs (e.g. due to shipping costs)
 - marginal costs $\neq$ replacement costs (e.g. due to shelving costs)

High income households pay higher markups



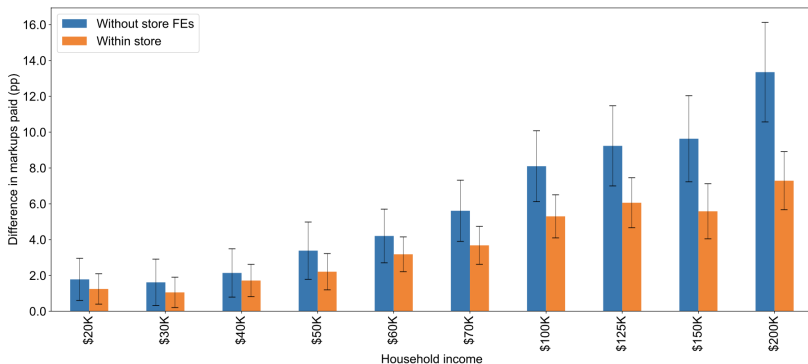
Within county markup gaps

$$\text{Markup}_{i,g,t} = \sum_{\ell} \beta_{\ell} 1\{i \text{ has income level } \ell\} + \gamma' X_i + \delta_{\text{County}} + \epsilon_{i,g,t}$$



Within store markup gaps robustness

$$\text{Markup}_{i,g,t} = \sum_{\ell} \tilde{\beta}_{\ell} 1\{i \text{ has income level } \ell\} + \tilde{\gamma}' X_i + \alpha_{\text{Store}} + \epsilon_{i,g,t}$$

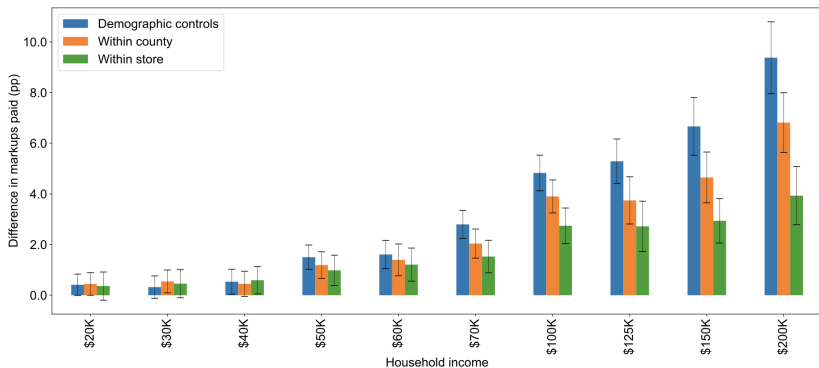


Markup differences for the same product

$$\text{Markup}_{i,g,t} = \sum_{\ell} \beta_{\ell} 1\{i \text{ has inc. level } \ell\} + \gamma' X_i + \psi_g + \epsilon_{i,g,t}$$

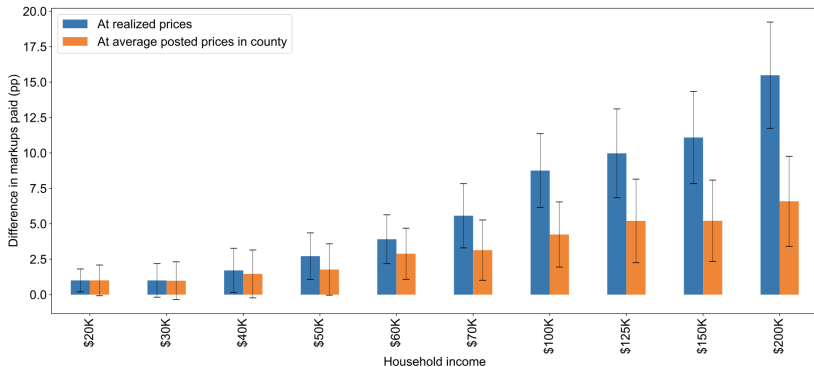
$$\text{Markup}_{i,g,t} = \sum_{\ell} \beta_{\ell} 1\{i \text{ has inc. level } \ell\} + \gamma' X_i + \psi_{\text{County}, g} + \epsilon_{i,g,t}$$

$$\text{Markup}_{i,g,t} = \sum_{\ell} \beta_{\ell} 1\{i \text{ has inc. level } \ell\} + \gamma' X_i + \psi_{\text{Store}, g} + \epsilon_{i,g,t}$$

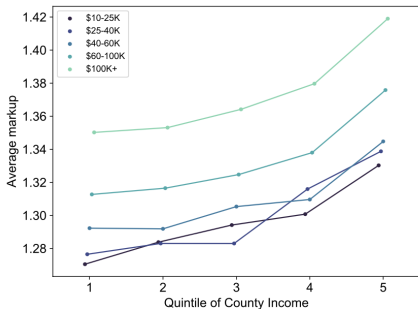


Markups differences: role of basket composition

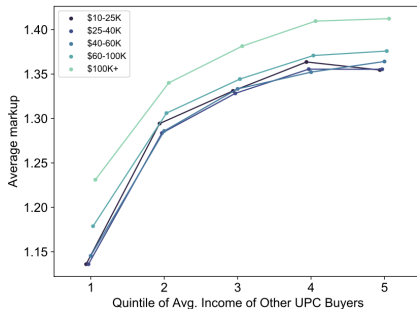
$$\text{Markup}_{i,g,t} = \sum_{\ell} \beta_{\ell} 1\{i \text{ has inc. level } \ell\} + \epsilon_{i,g,t}$$



Other buyers' income effect on markups



(a) County income



(b) Income of UPC's other buyers

Estimated spillovers of other buyers' incomes

<i>Log Retail Markup</i>	(1)	(2)	(3)
Log CBSA Income	0.071** (0.013)		
Log Income at Retailer's Locations		0.068** (0.030)	
Log Income of Other UPC Buyers			0.142** (0.038)
Household $\times$ Income Level in Year FEs	Yes		
Year FEs	Yes		
Household-Year FEs		Yes	Yes
Store FEs	Yes	Yes	
Store County-Year FEs		Yes	
Store-Year FEs			Yes
<i>N</i> (millions)	91.9	50.8	97.0
<i>R</i> ²	0.19	0.21	0.21

Macro elasticities across space vs over time

Outcome Sample	<i>Log Markup</i> 2006–2012		<i>Log Avg. Unit Price</i>			
			2004–2019		2004 and 2019	
	(1)	(2)	(3)	(4)	(5)	(6)
Log CBSA Income	0.104** (0.017)	0.099** (0.026)	0.127** (0.026)	0.128** (0.030)	0.097** (0.045)	0.096* (0.052)
Year FEs	Yes	Yes		Yes		Yes
CBSA FEs		Yes	Yes		Yes	
Year-Product Module FEs			Yes		Yes	
CBSA-Product Module FEs				Yes		Yes
<i>N</i> (millions)	133	133	18.3	18.3	2.2	2.2
<i>R</i> ²	0.03	0.04	0.99	0.99	0.99	0.99

Model

Households

- Each household i purchases goods $k = 1, \dots, K$:

$$u(\{c_{ik}\}) = \left(\sum_{k=1}^K (\beta_{ik} c_{ik})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

- Households know distribution of prices $F_k(p)$, but do not know which retailer sells at which price
- $\{q_{ik,n}\}_{n=1}^{\infty}$ probability that household i observes n price quotes for good k
- Households buy from the firm with the lowest price quote received if $p \leq R$

Search function and maximization problem

- $\mathcal{S} : s \rightarrow \{q_n\}_{n=1}^{\infty}$ and $Q_n(s) = \sum_{i=1}^n q_i$. Assume:
 - $Q_1(s)$ strictly decreasing in s with $\lim_{s \rightarrow 0} Q_1(s) = 1$ and $\lim_{s \rightarrow \infty} Q_1(s) = 0$
 - $Q_n(s)$ weakly decreasing in s and C^∞ for all n
- $p_{ik} = p(s_{ik}, F_k) = \sum_{n=1}^{\infty} q_{ik,n}(s) \mathbb{E}(p|n)$
- Each household i solves:

$$\max_{l_i, \{c_{ik}, s_{ik}\}} u(\{c_{ik}\}) \quad \text{s.t.} \quad \begin{cases} \sum_k t_i(c_{ik}, s_{ik}) + l_i = 1 & \text{(time const.)} \\ \sum_k p_{ik} c_{ik} = z_i l_i & \text{(budget const.)} \end{cases}$$

- where $t_i(c, s) = \frac{c}{a_i} s$

Optimality and aggregate search

- Households' optimal search level:

$$-\frac{\partial p_{ik}(s_{ik}, F_k)}{\partial s_{ik}} = \frac{z_i}{a_i} \equiv \phi_i$$

- Households indexed by $z \in (0, \infty)$, with cdf $H(z)$. Assume $\phi(z) = z/a(z)$
- Distribution of buyers' income for good k :

$$d\Lambda_k(z) = \frac{c_k(z)}{C_k} dH(z)$$

- Aggregate search behaviour for good k can be summarized by $\{\bar{q}_{k,n}\}_{n=1}^{\infty}$ where:

$$\bar{q}_{k,n} = \int_0^{\infty} q_{k,n}(z) d\Lambda_k(z) \quad \text{for all } n$$

Firms

- Good k supplied by M_k identical firms, CRS technology in labor:

$$\max_p \pi(p) = (p - 1)D_k(p)$$

- Define a *price-dispersed equilibrium* (Burdett and Judd (1983)).
The distribution of prices F is such that:
 - all firms choosing $p \in \text{supp}(F)$ make identical profits
 - charging $p \notin \text{supp}(F)$ results in lower profits

Equilibrium price distribution

- Given $\{\bar{q}_n\}_{n=1}^{\infty}$ with $\bar{q}_1 \in (0, 1)$ the unique equilibrium price distribution $F(p)$ is:

$$F(p) = \begin{cases} 0 & \text{if } p < \underline{p} \\ 1 - \Psi \left[\left(\frac{R-1}{p-1} \right) \bar{q}_1 \right] & \text{if } \underline{p} < p \leq R \\ 1 & \text{if } p > R \end{cases}$$

where:

- $\underline{p} = 1 + \frac{\bar{q}_1}{\sum_{n=1}^{\infty} n \bar{q}_n} (R - 1)$
 - $\Psi(\cdot)$ is the inverse of $y(x) = \sum_{n=1}^{\infty} n \bar{q}_n x^{n-1}$
- Firms pay entry cost f_e . Free entry determines M_k . In equilibrium:

$$\pi(p) - f_e = 0 \quad \forall p \in [\underline{p}, R]$$

Equilibrium

An equilibrium is a tuple $(F_k, \{c_k(z), s_k(z)\}_{z=0}^{\infty}, M_k)_{k=1, \dots, K}$ s.t.

- 1 chosen $c_k(z), s_k(z)$ for each household, for each product maximize household's utility
- 2 distribution of prices for good k , F_k , is a price dispersed equilibrium given aggregate search behaviour $\{\bar{q}_{k,n}\}_{n=1}^{\infty}$
- 3 M_k is such that firms make zero profits net of entry costs
- 4 resource constraints satisfied

Search and markups paid in the cross-section

- **Assumption 1:** for any non-degenerate distribution F , the mapping $\mathcal{S} : s \rightarrow \{q_n\}_{n=1}^{\infty}$ satisfies:

$$\sum_{n=1}^{\infty} \frac{d^2 Q_n}{ds^2} [\mathbb{E}(p|n) - \mathbb{E}(p|n+1)] > 0$$

- $p_{ss} > 0$
 - $\phi = -p_s(s, F) \Rightarrow \partial s / \partial \phi < 0$
- **Lemma 1:** Suppose Assumption 1 holds. If $\phi(z)$ is increasing in z , then:
 - $s_k(z)$ for any good k is decreasing in z
 - $p_k(z)$ and $\mu_k(z)$ for any good k are increasing in z .

Strategic interactions and aggregate markup

- **Lemma 2:** There exists some ϕ^{cutoff} such that if $\phi(z) < \phi^{\text{cutoff}}$ for all z then:
 - a stable dispersed-price equilibrium exists
 - in that equilibrium, household search decisions are strategic substitutes: $\partial s_{ik} / \partial s_{jk} \leq 0$ for $i \neq k$ in market k
- **Lemma 3:** In equilibrium:

$$\bar{\mu}_k := \frac{\int_{\underline{p}_k}^R p D_k(p) M_k dF_k(p)}{\int_{\underline{p}_k}^R D_k(p) M_k dF_k(p)} = 1 + (R - 1) \bar{q}_{k,1}$$

$$\bar{\mu} = 1 + (R - 1) \frac{\sum_{k=1}^K C_k \bar{q}_{k,1}}{\sum_{k=1}^K C_k}$$

- **Proposition 1** (*first order shift*): $K = 1$, Assumption 1, $\Lambda(z)$ initial distribution of buyers' income.

Consider a perturbation to $\Lambda(z)$ s.t. $\tilde{\Lambda}(z)$ FOSD $\Lambda(z)$.

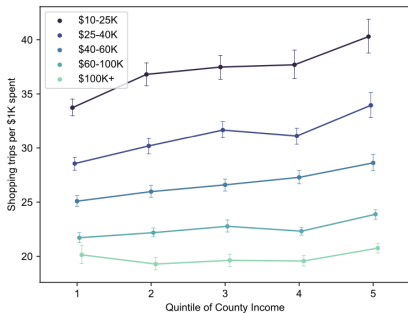
This perturbation leads to an increase in $\bar{\mu}$ if $\phi(z)$ increasing in z .

- **Proposition 2** (*mean-preserving spread*): $K = 1$, Assumption 1-2, $\Lambda(z)$ initial distribution of buyers' income.

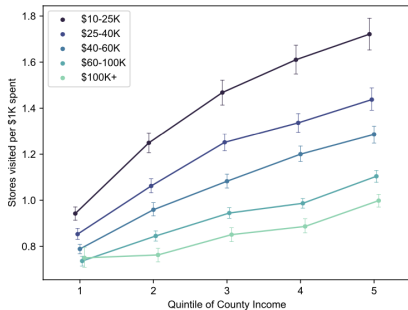
Consider a perturbation to $\Lambda(z)$ s.t. $\tilde{\Lambda}(z)$ is a mean preserving spread of $\Lambda(z)$.

This perturbation leads to an increase in $\bar{\mu}$ if $\phi(z)$ is increasing and convex in z .

Evidence on search behaviour



(a) Shopping trips per \$1K expenditures.



(b) Unique stores visited per \$1K expenditures.

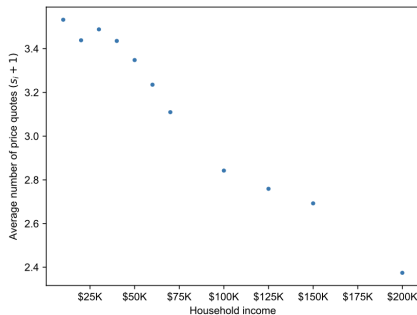
Calibration

Parameter		Value	Source
Number of products	K	10^+	Increasing $K > 10$ does not change results
Elasticity of substitution	σ	1^+	Cobb-Douglas
Quality shifters	$\beta_k(z)$	-	Match spending shares
Unit wage	w	1	Numeraire
Reservation price	R	3.3^+	98th percentile of markups in the data
Search mapping	$\mathcal{S}$	Poisson	Albrecht et al. (2021), Menzio (2021)
Opp. costs of search	$\phi(z)$	-	Match avg. markup paid by income group
Search productivity	$a(z)$	-	Solved from $\phi(z) = z/a(z)$
Household distribution	$H(z)$	-	Saez and Zucman (2019)

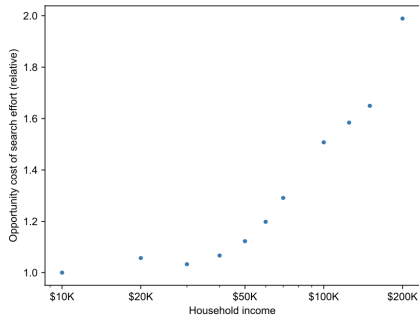
Search map functional form:

$$q_{ik,n+1} = \frac{s_{ik}^n \exp(-s_{ik})}{n!}$$

Calibrated statistics

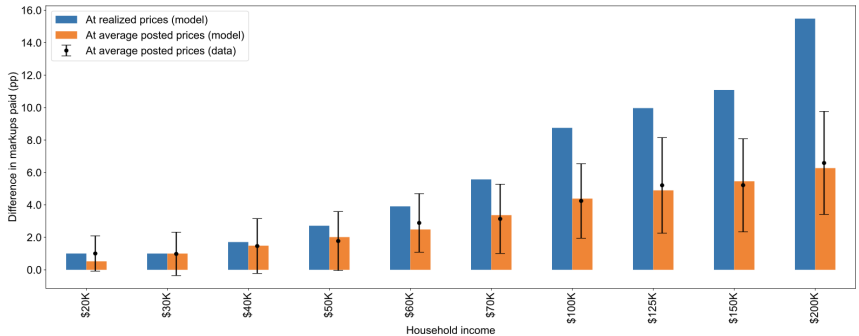


(a) Avg. price quotes received ($s(z) + 1$).

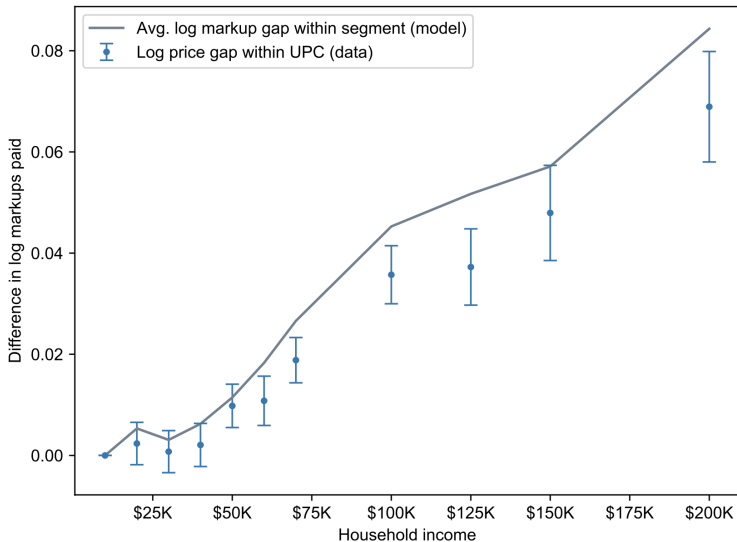


(b) Opportunity cost of search effort $\phi(z)$.

Contribution of basket composition to markup gap



Contribution of search effort to markup gap



Spillovers and strategic interaction

- Simulate the model using income distribution of 881 CBSAs and run regressions of:
 - markups paid and quotes received per purchase (proxy for search intensity)
 - on own income and average CBSA income

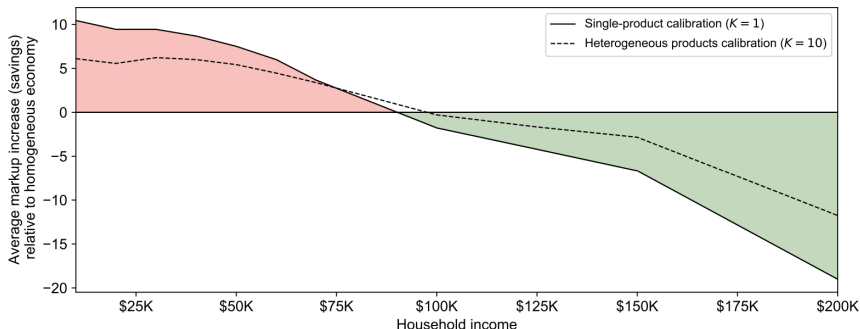
	<i>Log markup</i>			<i>Search intensity</i>		
	OLS	IV	Model	OLS	IV	Model
Log Own Income	0.032	0.054	0.032	-0.26	-0.40	-0.11
Log Others' Income	0.102	0.089	0.071	0.03	0.09	0.03

Model implications

Spillovers across income groups

Compare:

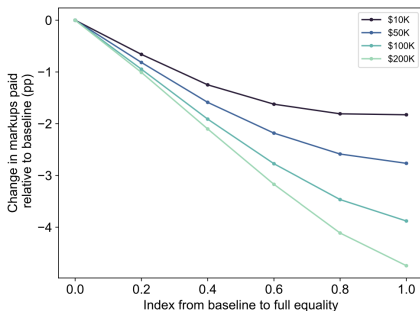
- Aggregate markup paid by each income group z in baseline calibration
- Aggregate markup in a representative household economy with income z and search productivity $a(z)$



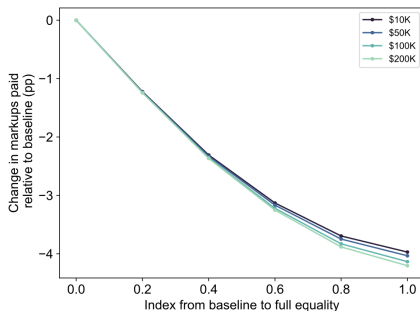
Effects of income dispersion

- Simulate the model under counterfactual household income distributions, scaling all incomes towards average per-capita income:

for each $\chi \in [0, 1]$
$$\tilde{H}(z) = H\left(\frac{z - \chi\bar{z}}{1 - \chi}\right)$$



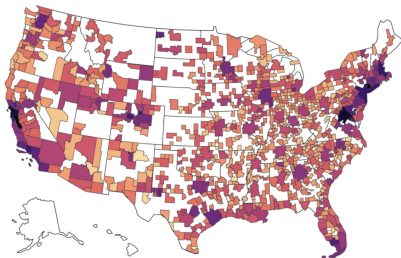
(a) Baseline model ($K = 10$).



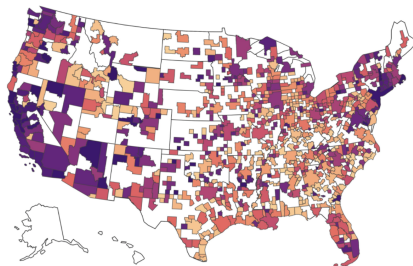
(b) Single-good model ($K = 1$).

Markups across cities (1)

a) predicted markups across CBSAs



b) markups across CBSAs in the data

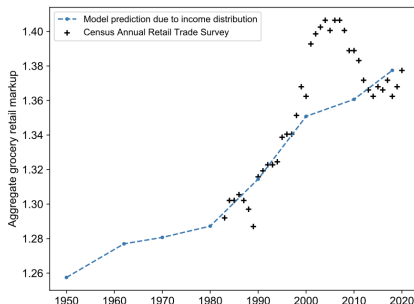
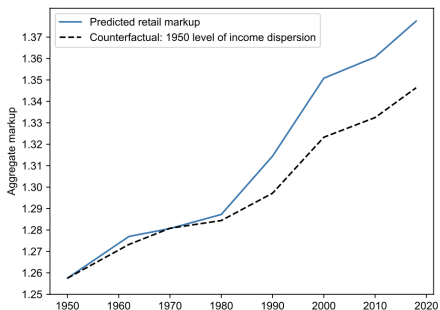


Markups across cities (2)

<i>Log CBSA Markup</i>	Model-Predicted		Data				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log CBSA Income	0.102** (0.001)	0.096** (0.002)	0.110** (0.006)	0.102** (0.007)			0.018 (0.015)
Gini Index		0.113** (0.014)		0.153** (0.057)			0.067 (0.058)
Log Model-Predicted Markup					1.056** (0.053)		0.824** (0.143)
Log Nested CES Markup						-0.720** (0.072)	-0.122* (0.074)
<i>N</i>	881	881	881	881	881	881	881
<i>R</i> ²	0.84	0.85	0.27	0.28	0.31	0.10	0.31

Changes in income distribution from 1950-2018

- **Assumption:** For every income distributions in 1950-2018 household with income z have same $a(z), \phi(z)$ as in 2007 data
 - Motivation: elasticities of markups to aggregate income nearly identical in time series and cross-section

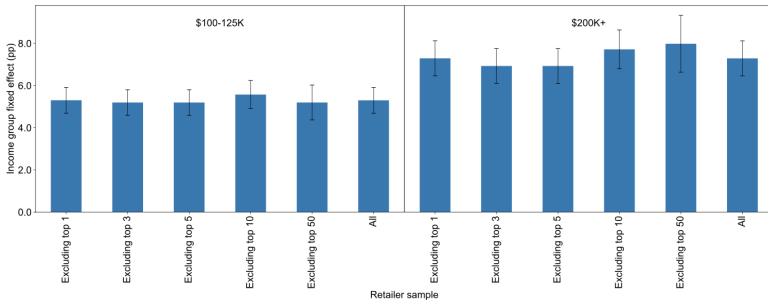


Conclusion

- Income level/inequality are important determinants of markups:
 - High income households exert less search effort, buy higher share of high-markup goods
 - Firm markups are determined by the distribution of their customers' incomes
- Developed a model which accounts for this micro-evidence to show:
 - how changes in income level/inequality shape retail markups
 - markups in the data exhibit variation across cities
 - changes in the income distribution in 1950-2018 can account for a substantial rise in retail markups

Appendix

Robustness of markup gap

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<i>Markup gap (pp) relative to <\$20K</i>	Demographics		Within County		Within Store	
	\$100K	\$200K	\$100K	\$200K	\$100K	\$200K
Baseline	10.8	17.7	8.8	13.9	5.3	7.3
Weighting by sales	10.4	17.8	7.8	12.7	4.5	6.1
Using PromoData base price	9.7	16.0	8.0	12.7	5.6	7.9
Using PromoData market-level price	8.5	16.3	6.6	11.7	5.8	8.9
With day-of-week fixed effects	10.6	17.5	8.6	13.8	5.1	7.1
With supply-side controls	10.6	17.4	8.5	13.6	5.2	7.2
Excluding perishable items	10.5	17.1	8.4	13.4	4.8	6.7

- **Assumption 1:** for any non-degenerate distribution F , the mapping $\mathcal{S} : s \rightarrow \{q_n\}_{n=1}^{\infty}$ satisfies:

$$\sum_{n=1}^{\infty} \frac{d^2 Q_n}{ds^2} [\mathbb{E}(p|n) - \mathbb{E}(p|n+1)] > 0$$

- **Assumption 2:** for any non-degenerate distribution F , the mapping $\mathcal{S} : s \rightarrow \{q_n\}_{n=1}^{\infty}$ satisfies:

$$\sum_{n=1}^{\infty} \left(\frac{d^2 Q_1}{ds^2} \frac{d^2 Q_n}{ds^2} - \frac{dQ_1}{ds} \frac{d^3 Q_n}{ds^3} \right) [\mathbb{E}(p|n) - \mathbb{E}(p|n+1)] \geq 0$$

Household density $H(z)$ over time

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Constructed from Saez, Zucman (2019)

